Prospectus:
How do context and time modulate the geometric trajectory of episodic memory?

Paxton Fitzpatrick

February 6, 2019

1 Rationale

In traditional free recall studies, participants are presented with a series of items (e.g. words, pictures, etc.), then asked to recall them to the best of their abilities. The behavioral findings yielded by such studies on recall order and transitions (Murdock, 1962; Kahana, 1996), temporal and semantic clustering tendencies (Howard and Kahana, 2002b), and response latency (Murdock and Okada, 1970) have given rise to numerous models of memory encoding, storage, and retrieval (e.g., Atkinson and Shiffrin, 1968; Howard and Kahana, 2002a; Sederberg et al., 2008). The functional predictions of these models have, in turn, spurred investigations into their biological underpinnings (e.g., Polyn et al., 2005; Manning et al., 2011, 2012; Howard et al., 2012). However, the free recall paradigm is limited in its ability to assess episodic memory for real-world events (see Koriat and Goldsmith, 1994; Chen et al., 2017; Huk et al., 2018; Baldassano et al., 2018; Heusser et al., 2018a; Manning, 2019). Its inherently discretized stimuli, binary accuracy evaluation, and reliance on participants' precise reproduction of presented stimuli clash with the steady, continuous evolution of real-life experiences and the value we instead place on capturing their general gist in assessing our (or others') memories for them.

The emergence and popularization of naturalistic paradigms offers the opportunity to create and refine new context-driven models of episode perception and memory that better reflect the use of these systems in our day-to-day lives. In turn, behavioral predictions stemming from such improved models better inform and constrain attempts to identify underlying structures and networks. For example, a number of these models (e.g., Chen et al., 2017) have motivated imaging studies that have recovered neural representation of ongoing events at multiple, hierarchical timescales, largely based in networks of regions including the medial prefrontal cortex (mPFC), medial temporal lobe (MTL), and anterior cingulate cortex (ACC). These multi-timescale episodic memory findings join similar discoveries for reinforcement learning (e.g., Bernacchia et al., 2011; Bromberg-Martin et al., 2010), motor learning (e.g., Fusi et al., 2007; Kim et al., 2015), and spatial memory (e.g., Strange et al., 2014) in support of the long-standing challenge against the functional separation between memory and ongoing information processing (see Hasson et al., 2015). Yet, because memory for an experience is rarely as detailed as its initial perception, there must exist a mechanism for transforming and compressing neural representations of experiences' content after they are initially perceived.

The rich complexity and fluidity of real-world experiences (as well as the numerous idiosyncratic ways they may be later described) make their content far more difficult to quantify and analyze than the simple stimuli and responses native to list-learning studies. Previous naturalistic paradigms attempting to investigate multi-timescale representations of episode perception and memory have had to approximate these experiences by either synthetically regulating their temporal unfolding (e.g., Lerner et al., 2011) or disregarding their content altogether and using state changes in shared neural responses to infer event transitions (e.g., Howard et al., 2014).

Heusser et al. (2018a) recently developed a model that can capture the explicit moment-by-moment content of a naturalistic experience. The framework portrays the content's dynamic evolution as a geometric trajectory through a self-defined representational space. Verbal recollections of the experience may then be similarly cast as "thought trajectories" that traverse the same space. By assessing the similarity between moments of content and subsequent recountings along their trajectories' various axes, this model disentangles the overall *quality* of an episodic memory from limiting "proportion recalled" measures and enables identification of event boundaries based on the explicit content of experiences and memories.

Predictions from Heusser et al. (2018a)'s model have been successfully leveraged to implicate networks of brain regions in processing the geometric structure of an ongoing experience at multiple timescales (e.g., bilateral frontal cortex, cingulate cortex) and subsequently transforming it into compressed memory (e.g., vmPFC, rMTL, ACC), adding a level of refinement to Baldassano et al. (2017)'s findings. However, a persisting limitation to these findings is their assessment of the experiences they study in isolation from all other experiences. Our ongoing perception and memory is shaped by physical and mental contexts extending well beyond the length of a single experiment session. Heusser et al. (2018a)'s content-sensitive model enables examination of interactions between multiple activations of these processing and compression networks across time. My thesis will investigate how the geometric structure of episodic memory is modulated by factors including the passage of time, contextual congruence of existing memories, and accuracy in predicting future experiences. Additionally, I plan to link my findings with established anatomical and physiological results to inform and constrain hypotheses for investigations into these interactions' neurobiological bases.

2 Methods

2.1 Experimental design and data collection

Participants ($n = 60$) are divided into two equal groups. Both groups attend two study sessions, six to eight days apart. In the first session, all participants view and immediately recount the pilot episode of *Atlanta* to the best of their abilities. Participants then predict how the second episode will unfold the following week. In the second session, both groups begin by recounting the prior week's episode. Then, group 1 views and recounts the second episode of *Atlanta*, while group 2 views and recounts the pilot episode of *Arrested Development*. All participant audio is recorded using the viewing computer's built-in microphone and automatically transcribed using Google Cloud Speech-to-Text.

2.2 Stimulus Annotation

To generate the content corpus for each stimulus video, I use `ANVIL` (Kipp, 2001), an audiovisual annotation tool, with a custom XML specification file to record the following features for each camera shot:

- A brief description of physical occurrences and characters' actions
- A brief description of characters' emotional states and various subtextual implications
- A list of characters appearing on-screen
- The presence or absence of music
- A transcription of all dialogue
- The name of the character speaking
- Whether the shot took place indoors or outdoors
- The shot's setting

2.3 Dynamic content modeling, event segmentation, and video-recall mapping

Following Heusser et al. (2018a), I use `Hypertools` (Heusser et al., 2018b) to train a topic model (Blei et al., 2003) on the video annotations, creating a high-dimensional space whose axes represent the latent themes (or "topics") in the episode's content. I then use this model to transform both the video annotations and each participant's recall and prediction transcripts into this common topic space. Within this space, the video, recalls, and predictions form trajectories along which the coordinates at each point reflect the weighted mixture of topics represented in the content at that point. This allows for comparing between them in a way that is insensitive to specific word choice. I then use linear interpolation to resample the video trajectory to a consistent one-second resolution and dynamic time warping (Berndt and Clifford, 1994) to standardize the time dimension of the recall and prediction trajectories, separately.

Next, I divide the continuous video, recall, and prediction trajectories into discrete "events." Using Hidden Markov models (Rabiner, 1989), I parse the trajectories' timeseries into K (non-repeating) states, such that the distribution of topics represented at each moment were maximally similar within events, while maximally dissimilar across events. To map between viewed and remembered/predicted events, I compute the average topic vector for each recalled/predicted event and assigned it to stimulus event whose average topic vector is maximally similar (i.e., closest to it in topic space).

3 Analyses to be employed and expected results

3.1 Comparing immediate and delayed recall

To investigate how the one-week delay period alters the spatiotemporal unfolding of an episodic memory, I plan to examine differences in both the across-subjects average and individual participants' recall trajectories on a full-model scale, as well as on a per-event basis. To achieve the first, I will examine how this delay period affects the general figure of retrieved content from session 1's episode. Projecting the content models down into a shared 2D plane reveals an episode-unique unique geometric shape for each stimulus that is traversed by the shapes of participants' recall models (see Heusser et al., 2018a for specific details). Previously, average 2D-embedded recall shapes have shown near-perfect recovery of the video content, with individuals' shapes displaying idiosyncratic differences. I plan to measure the change in the variation among individuals' recall geometries with the expectation that they will converge and "smooth" as participants retain the broad plotline of the episode and discard minor details.

Additionally, I can asses changes in both the proportion of remembered events and the quality of memory for events remembered in both recall phases. To do this, I will examine whether the Hidden Markov model identifies fewer states for in the delayed models, as well as quantify changes in the topic space-distance between remembered events and corresponding presented events. I can then examine the features shared by video events whose memory representations were least blurred with time to infer features that contribute to successful maintenance in episodic memory.

3.2 Measuring the effect of contextual familiarity on future memory success

I also intend to explore how three progressive levels of memory processing of prior experiences modulate memory for subsequent ones. First, I will examine the effect simple exposure to a situation has on later encoding and retrieval of similar contexts. Most broadly, I can analyze differences in memory performance during session 2 between participants in my "contextually congruent condition" (i.e. those who viewed the second *Atlanta* episode during session 2) and my "contextually incongruent condition" (i.e. those who instead viewed *Arrested Development*) on both a full-model scale and per-event basis. More finely, by defining a new topic space shared by mulstiple stimuli, I can compute their similarities at each moment in time and analyze whether events viewed during session 2 were remembered more frequently or more fully as a function of thier similarity to session 1 events.

I can then perform a similar set of analyses on the effect that successfully transforming experiences into compressed memories has on subsequent iterations of the same process, as well as how successfully reinstating various contextual features affects subsequent memory encoding. In the case of the former, I will relate metrics of topical distance between participants' immediate recalls in session 1 and episode viewed during session 2 to memory quality for session 2. For the latter analysis, I would perform a similar set of operations using participants' delayed recalls in place of their immediate recalls.

3.3 Relating prediction ability to memory success

I hope as well to determine how the ability to accurately predict a future experience impacts its eventual encoding, storage, and retrieval. This may be investigated both on a full model-scale and event-wise basis. First, I will quantify overall prediction accuracy based on models of group 1 participants' prediction transcripts, projected into a topic space defined by the content of the second *Atlanta* episode. In this way, the prediction trajectory is functionally equivalent to a recall trajectory for an experience that has yet to occur. I can then ask whether measures of group 1's recall accuracy in session 2 fluctuate as a function of their prediction accuracy (i.e., the moments-wise distance between the two trajectories in topic space). Second, I segment participants' prediction models into discrete events and determine the topic distributions in each. I can then ask whether memory for events in episode 2 with similar thematic elements improves as a result.

4 Implications

An asset to Heusser et al.'s (2018a) framework is the identification of event boundaries from the content/recall models' temporal correlation matrices, as these matrices hold the combined representation of events at all timescales. I hope to find changes in the proportional emphasis on different timescales between immediate and delayed recall to form a basis for an imaging study to classify activation patterns representing these various timescales and attempt to identify their neural bases.

With the analyses relating contextual similarity to memory success, I hope to first contribute support to one of two classes of competing models. Interference theory-based models (see Wixted and Rohrer, 1993) would propose a decline in accuracy for contextually similar events, while schema- or scaffolding-based models (see Bobrow and Norman, 1975) would suggest the opposite. Both theories have informed numerous successful studies on their biological backing (e.g., Mecklinger et al. 2003; Jonides and Nee 2006; Lyon et al. 2008 and Castel 2005; van Kesteren et al. 2012; Gilboa and Marlatte 2017, respectively). Discovering a behavioral pattern change resulting from each of the three levels of memory processing segregated by my model will guide future imaging work to search for corresponding neural pattern changes in an analogous style to Baldassano et al. (2017) and Heusser et al. (2018a).

Finally, prior work (e.g., Polyn et al., 2009; Manning et al., 2011) has suggested that MTL and mPFC play a roll in representing various contextual features of ongoing experiences and retrieving them during recall. Additionally, other work implicates mPFC in processing sensitivity to experiences that mismatch expectations (e.g., Richards et al., 2014). I hope for model to yield insight into what features of ongoing experiences, existing memories, and successful predictions do and do not contribute to successful remembering. These results will guide stimulus design for a series of imaging studies based on heavy variation of a single feature at a time, in an attempt to discover their specific neural representations.

References

Atkinson, R. C. and Shiffrin, R. M. (1968). Human memory: A proposed system and its control processes. In Spence, K. W. and Spence, J. T., editors, *The psychology of learning and motivation*, volume 2, pages 89–105. Academic Press, New York.

Baldassano, C., Chen, J., Zadbood, A., Pillow, J. W., Hasson, U., and Norman, K. A. (2017). Discovering event structure in continuous narrative perception and memory. *Neuron*, 95(3):709–721.

Baldassano, C., Hasson, U., and Norman, K. A. (2018). Representation of real-world event schemas during narrative perception. *Journal of Neuroscience*, 38(45):9689–9699.

Bernacchia, A., Seo, H., Lee, D., and Wang, X. J. (2011). A reservoir of time constants for memory traces in cortical neurons. *Nature Neuroscience*, 14(3):366–72.

Berndt, D. J. and Clifford, J. (1994). Using dynamic time warping to find patterns in time series. In *KDD workshop*, volume 10, pages 359–370.

Blei, D. M., Ng, A. Y., and Jordan, M. I. (2003). Latent dirichlet allocation. *Journal of Machine Learning Research*, 3:993 – 1022.

Bobrow, D. G. and Norman, D. A. (1975). Some principles of memory schemata. In *Representation and understanding*, pages 131–149. Elsevier.

Bromberg-Martin, E., Matsumoto, M., Nakahara, H., and Hikosaka, O. (2010). Multiple timescales of memory in lateral habenula and dopamine neurons. *Neuron*, 67(3):499–500.

Castel, A. D. (2005). Memory for grocery prices in younger and older adults: The role of schematic support. *Psychology and Aging*, 20(4):718–721.

Chen, J., Leong, Y. C., Norman, K. A., and Hasson, U. (2017). Shared experience, shared memory: a common structure for brain activity during naturalistic recall shared experience, shared memory: a common structure for brain activity during naturalistic recall. *Nature Neuroscience*, 20(1):115–125.

Fusi, S., Asaad, W., Miller, E. K., and Wang, X. J. (2007). A neural model of flexible sensori-motor mapping: learning and forgetting on multiple timescales. *Neuron*, 54:319–333.

Gilboa, A. and Marlatte, H. (2017). Neurobiology of schemas and schema-mediated memory. *Trends Cogn Sci*, 21(8):618–631.

Hasson, U., Chen, J., and Honey, C. J. (2015). Hierarchical process memory: memory as an integral component of information processing. *Trends in Cognitive Science*, 19(6):304–315.

Heusser, A. C., Fitzpatrick, P. C., and Manning, J. R. (2018a). How is experience transformed into memory? *bioRxiv*, 409987.

Heusser, A. C., Ziman, K., Owen, L. L. W., and Manning, J. R. (2018b). HyperTools: a Python toolbox for gaining geometric insights into high-dimensional data. *Journal of Machine Learning Research*, 18(152):1–6.

Howard, M. W. and Kahana, M. J. (2002a). A distributed representation of temporal context. *Journal of Mathematical Psychology*, 46:269–299.

Howard, M. W. and Kahana, M. J. (2002b). When does semantic similarity help episodic retrieval? *Journal of Memory and Language*, 46:85–98.

Howard, M. W., MacDonald, C. J., Tiganj, Z., Shankar, K. H., Du, Q., Hasselmo, M. E., and H., E. (2014). A unified mathematical framework for coding time, space, and sequences in the medial temporal lobe. *Journal of Neuroscience*, 34(13):4692–4707.

Howard, M. W., Viskontas, I. V., Shankar, K. H., and Fried, I. (2012). Ensembles of human MTL neurons "jump back in time" in response to a repeated stimulus. *Hippocampus*, 22:1833–1847.

Huk, A., Bonnen, K., and He, B. J. (2018). Beyond trial-based paradigms: continuous behavior, ongoing neural activity, and naturalistic stimuli. *Journal of Neuroscience*, 10.1523/JNEUROSCI.1920-17.2018.

Jonides, J. and Nee, D. E. (2006). Brain mechanisms of proactive interference in working memory. *Neuroscience*, 139:181–193.

Kahana, M. J. (1996). Associative retrieval processes in free recall. *Memory & Cognition*, 24:103–109.

Kim, S., Ogawa, K., Lv, J., Schweighofer, N., and Imamizu, H. (2015). Neural substrates related to motor memory with multiple timescales in sensorimotor adaptation. *PLoS biology*, 13(12):e1002312.

Kipp, M. (2001). Anvil-a generic annotation tool for multimodal dialogue. In *Seventh European Conference on Speech Communication and Technology*.

Koriat, A. and Goldsmith, M. (1994). Memory in naturalistic and laboratory contexts: distinguishing accuracy-oriented and quantity-oriented approaches to memory assessment. *Journal of Experimental Psychology: General*, 123(3):297–315.

Lerner, Y., Honey, C. J., Silbert, L. J., and Hasson, U. (2011). Topographic mapping of a hierarchy of temporal receptive windows using a narrated story. *Journal of Neuroscience*, 31(8):2906–2915.

Lyon, D. R., Gunzelmann, G., and Gluck, K. A. (2008). A computational model of spatial visualization capacity. *Cognitive Psychology*, 57(2):122–152.

Manning, J. (2019). Episodic memory: mental time travel or a quantummemory wave'function? *PsyArXiv*.

Manning, J. R., Polyn, S. M., Baltuch, G., Litt, B., and Kahana, M. J. (2011). Oscillatory patterns in temporal lobe reveal context reinstatement during memory search. *Proceedings of the National Academy of Sciences, USA*, 108(31):12893–12897.

Manning, J. R., Sperling, M. R., Sharan, A., Rosenberg, E. A., and Kahana, M. J. (2012). Spontaneously reactivated patterns in frontal and temporal lobe predict semantic clustering during memory search. *The Journal of Neuroscience*, 32(26):8871–8878.

Mecklinger, A., Weber, K., Gunter, T. C., and Engle, R. W. (2003). Dissociable brain mechanisms for inhibitory control: effects of interference content and working memory capacity. *Brain Research Cognitive Brain Research*, 18(1):26–38.

Murdock, B. B. (1962). The serial position effect of free recall. *Journal of Experimental Psychology*, 64:482–488.

Murdock, B. B. and Okada, R. (1970). Interresponse times in single- trial free recall. *Journal of Verbal Learning and Verbal Behavior*, 86:263–267.

Polyn, S. M., Natu, V. S., Cohen, J. D., and Norman, K. A. (2005). Category-specific cortical activity precedes retrieval during memory search. *Science*, 310:1963–1966.

Polyn, S. M., Norman, K. A., and Kahana, M. J. (2009). A context maintenance and retrieval model of organizational processes in free recall. *Psychological Review*, 116(1):129–156.

Rabiner, L. (1989). A tutorial on Hidden Markov Models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2):257–286.

Richards, B., Xia, F., Santoro, A., Husse, J., Woodin, M., Josselyn, S., and Frankland, P. (2014). Patterns across multiple memories are identified over time. *Nature Neuroscience*, 17(7):981.

Sederberg, P. B., Howard, M. W., and Kahana, M. J. (2008). A context-based theory of recency and contiguity in free recall. *Psychological Review*, 115(4):893–912.

Strange, B., Witter, M., Lein, E., and Moser, E. (2014). Functional organization of the hippocampal longitudinal axis. *Nature Reviews Neuroscience*, 15(10):655.

van Kesteren, M. T. R., Ruiter, D. J., Fernández, G., and Henson, R. N. (2012). How schema and novelty augment memory formation. *Trends Neurosci*, 35(4):211–9.

Wixted, J. T. and Rohrer, D. (1993). Proactive interference and the dynamics of free recall. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 19:1024–1039.